Passwords Are Failing

How to Protect Yourself and Your Organization Using Password Managers and Passkeys

Whoami

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What is a Password?

How Many Passwords Do You Have?

The average person today has 100+ passwords

Raise your hand if…

- You use the **same password for multiple sites**
- You have **forgotten a password this week**
- You have **written a password somewhere**

Scenario

- An **attacker gets your password** from a data breach somewhere — maybe a shopping site.
- They try that same password on your email account.

- It works.

- Now they send an email from your account to finance saying:

- ‘Please urgently wire $50,000 to this vendor.

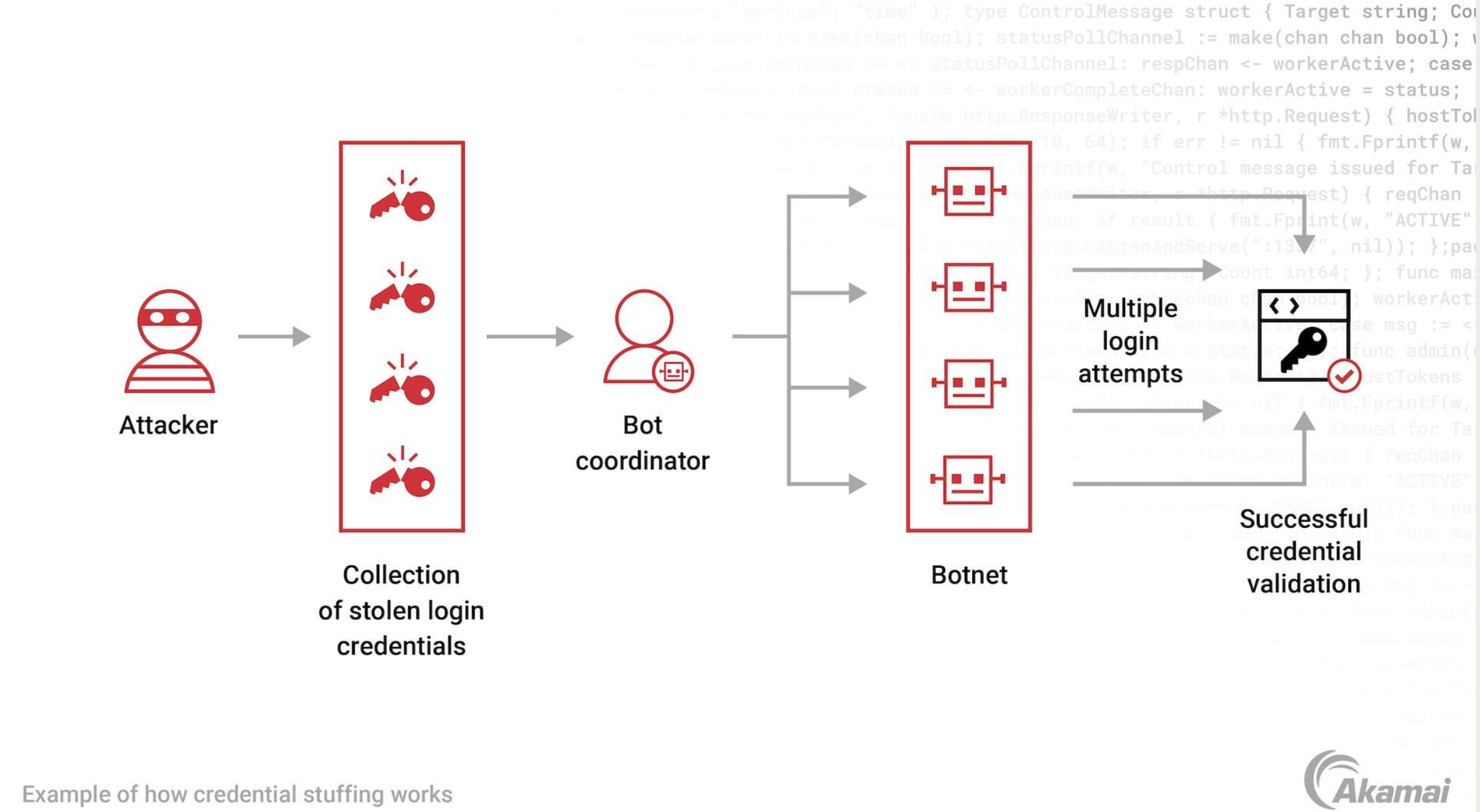
How many passwords were compromised?

- https://haveibeenpwned.com/
  - 959 Data Breaches
  - 17,501,153,109 Compromised Accounts – 17 Billion

- https://haveibeenpwned.com/passwords

What attackers do with these?

Credential Stuffing



Example of how credential stuffing works

⚠ THE CORE PROBLEM

Why Passwords Keep Failing Us

Passwords suffer from three fundamental weaknesses that make them unreliable as a security measure — and attackers know it.

1

People Reuse Them

The average person reuses the same password across multiple accounts, creating a chain of vulnerability.

2

Easy to Guess

Common patterns like names, dates, and dictionary words make passwords predictable to automated tools.

3

Stolen in Breaches

Data breaches expose millions of credentials. If your LinkedIn, Amazon, or Gmail password leaks, attackers try it **everywhere.**

BREACH DATA

Real-World Breaches at Massive Scale

Millions of passwords leak every year in high-profile breaches. Even if **your organization is secure**, attackers may already have your employees' passwords from somewhere else.

500M+

LinkedIn

Accounts compromised in one of the largest professional network breaches

3B

Yahoo

The single largest breach in history — every Yahoo account was affected

530M

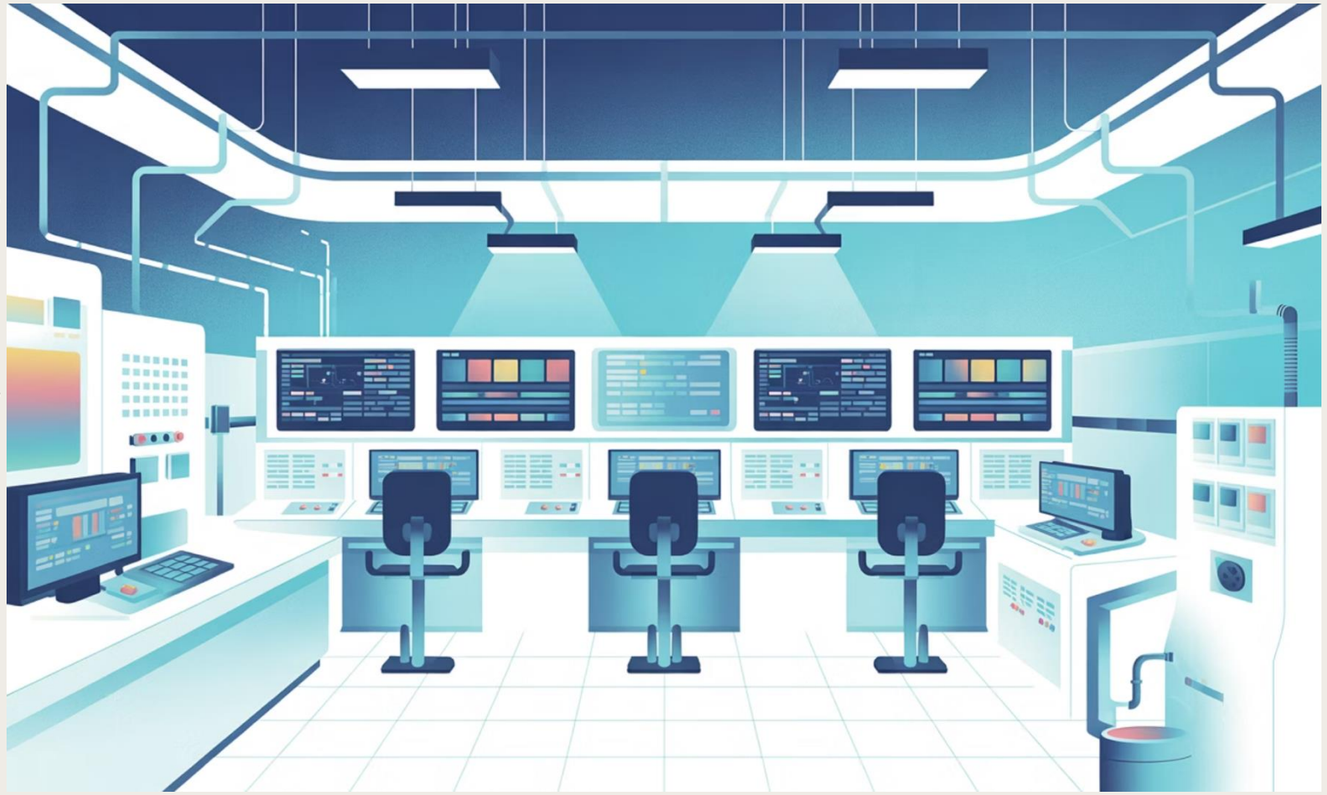
Facebook

User records exposed including phone numbers and personal data

⚠ WHY THIS MATTERS

Water Utilities Are Critical Infrastructure

When attackers compromise accounts at water utilities, the consequences ripple far beyond IT. A single breached credential can lead to devastating outcomes across your entire operation.

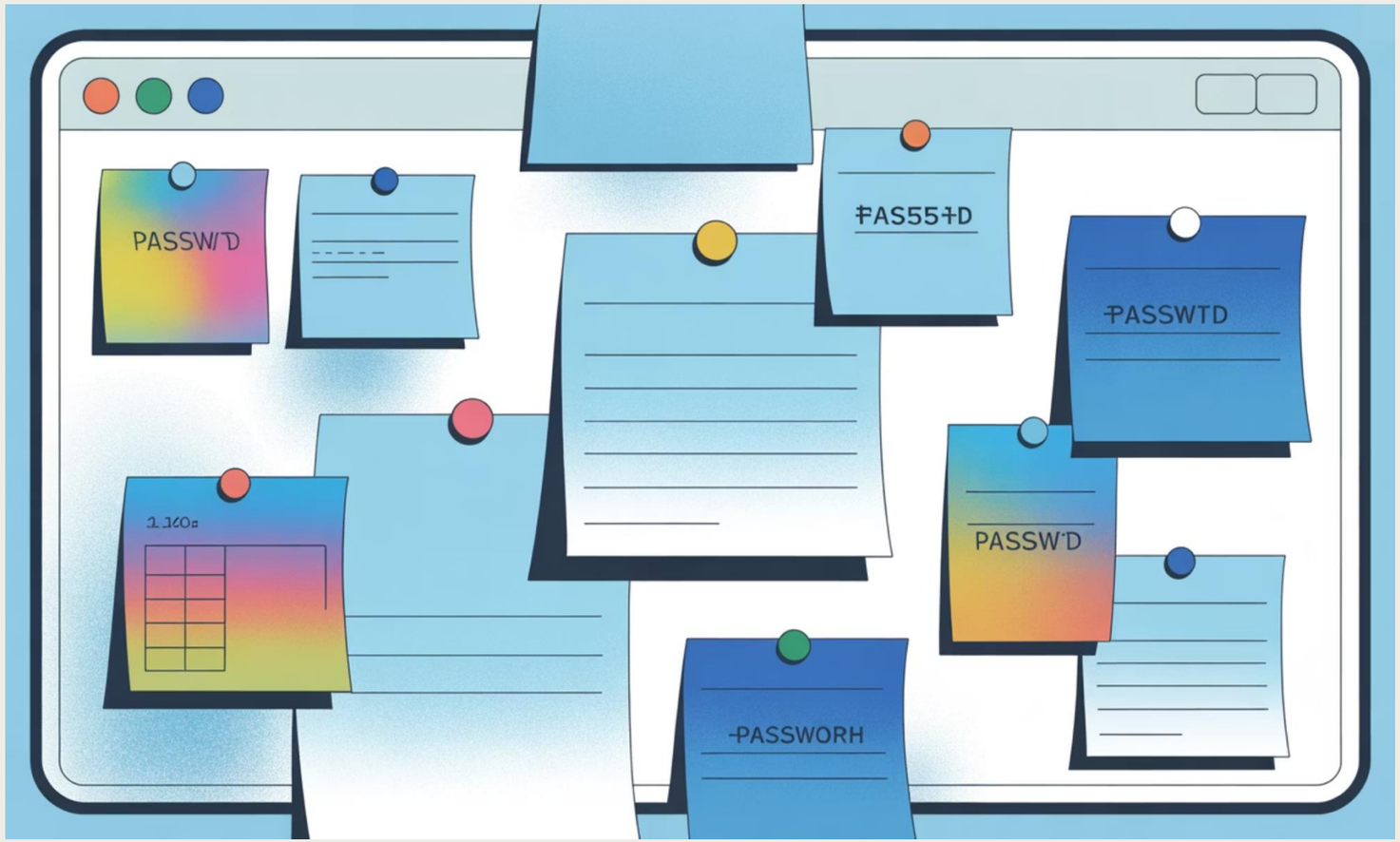


Disrupt Operations **Steal Sensitive Data** **Launch Ransomware** **Shut Down Systems**

The **#1 entry point** for attackers? **Passwords.**

THE HUMAN FACTOR

The Password Trap



What We're Told to Do

- Make passwords long and complex
- Make every password unique
- Change them frequently

What Actually Happens

Humans **cannot** remember 50+ complex passwords. So, they fall back on dangerous shortcuts:

- Sticky notes on monitors
- Same password everywhere
- Simple, predictable passwords

🛡 KNOW THE RISK

Weak Passwords Crumble in Seconds

❌ Bad Password

`Water123`

Cracked instantly by automated tools

⚠️ Still Bad

`WaterPlant2024!`

Predictable pattern — easily guessed

✅ Strong Password

`j7$kQ!9mXvL@2pNw`

Random, unique, stored in a password manager

The Speed of Modern Attacks

- Attackers use automated cracking tools that can try **millions of guesses per second**. Simple passwords — even those with a number and symbol — are no match.
- Common patterns like company names, years, and keyboard sequences are the first things these tools try.

BREACH DATA

Cracking the Passwords

Time it takes a hacker to brute force your password in 2025

Hardware: 12 x RTX 5090 | Password hash: bcrypt (10)

Number of Characters	Numbers Only	Lowercase Letters	Upper and Lowercase Letters	Numbers, Upper and Lowercase Letters	Numbers, Upper and Lowercase Letters, Symbols
4	Instantly	Instantly	Instantly	Instantly	Instantly
5	Instantly	Instantly	57 minutes	2 hours	4 hours
6	Instantly	46 minutes	2 days	6 days	2 weeks
7	Instantly	20 hours	4 months	1 year	2 years
8	Instantly	3 weeks	15 years	62 years	164 years
9	2 hours	2 years	791 years	3k years	11k years
10	1 day	40 years	41k years	238k years	803k years
11	1 weeks	1k years	2m years	14m years	56m years
12	3 months	27k years	111m years	917m years	3bn years
13	3 years	705k years	5bn years	56bn years	275bn years
14	28 years	18m years	300bn years	3tn years	19tn years
15	284 years	477m years	15tn years	218tn years	1qd years
16	2k years	12bn years	812tn years	13qd years	94qd years
17	28k years	322bn years	42qd years	840qd years	6qn years
18	284k years	8tn years	2qn years	52qn years	463qn years

Hive Systems

Read more and download at
hivesystems.com/password

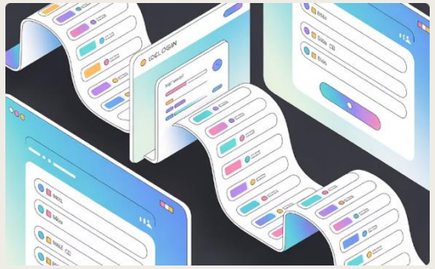
ATTACK METHODS

How Attackers Actually Break In

Most attacks **don't involve hacking systems** — they simply steal or guess passwords using well-known techniques.

Password Guessing

Automated tools cycle through common passwords and predictable patterns at incredible speed.



Credential Stuffing

Passwords stolen from one breach are tested across thousands of other sites and services automatically.

Phishing Emails

Fake login pages trick users into entering their real credentials, handing them directly to attackers.

THE SOLUTION

Password Managers: Your First Line of Defense

Instead of remembering dozens of passwords, use a **password manager** — a secure vault that handles the hard work for you. You only need to remember **one master password.**

Creates

Generates strong, random passwords for every account

Stores

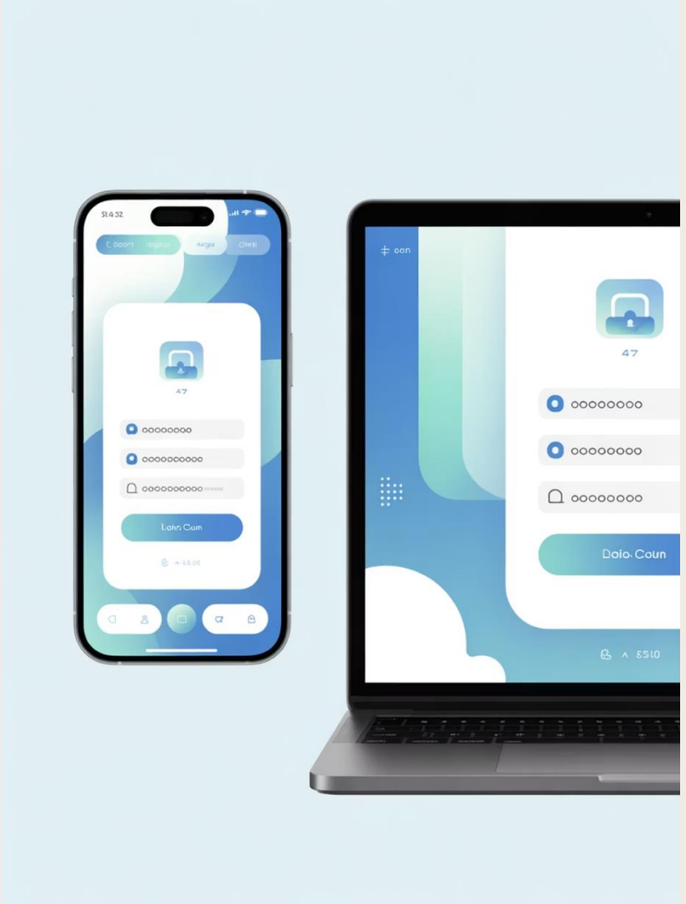
Encrypts and secures all credentials in one protected vault

Fills

Automatically fills login forms — no typing, no mistakes

TAKE ACTION

Choosing a Password Manager



Popular Options

1Password
Enterprise-ready with team sharing

Bitwarden
Open-source and highly trusted

LastPass
Widely adopted across organizations

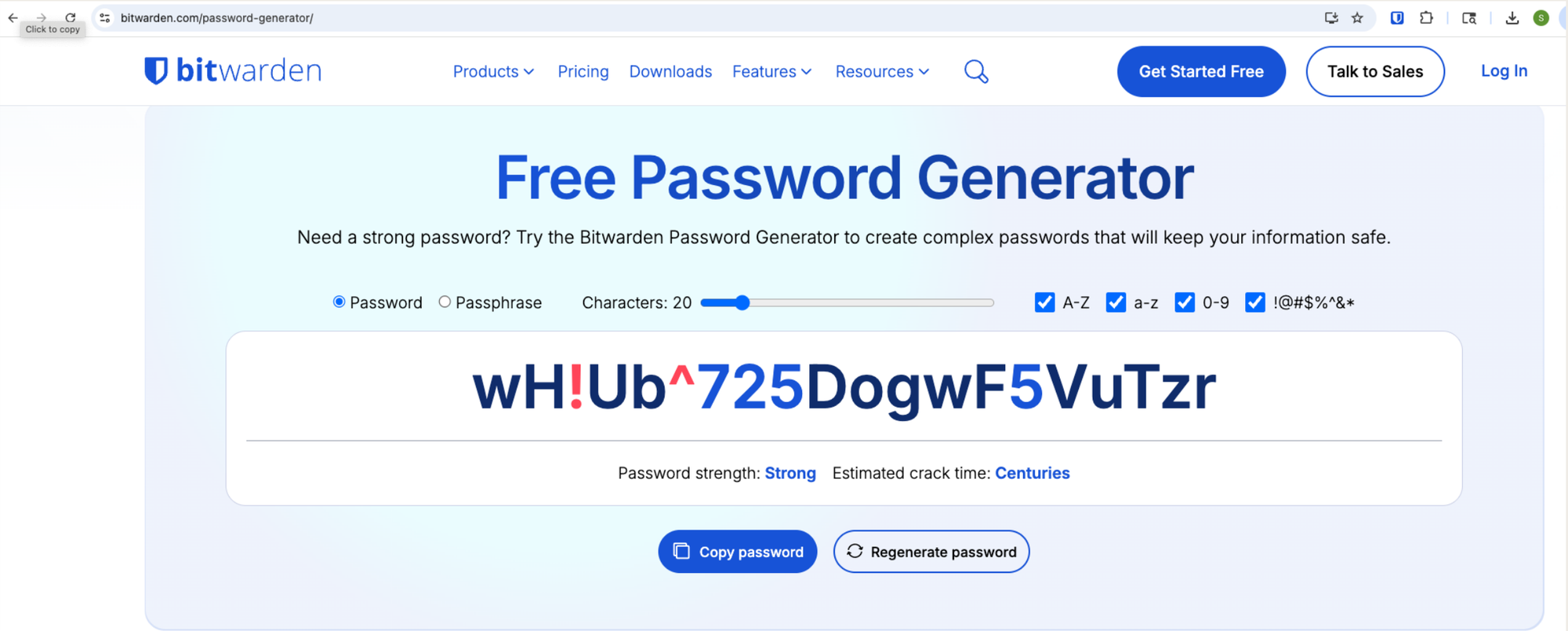
Dashlane
Built-in VPN and dark web monitoring

✓ Unique password for every site · ✓ No more reuse · ✓ Safer than sticky notes

TAKE ACTION

Password Manager - Demo

https://bitwarden.com/



🔑 PASSWORD STRENGTH

What a Strong Password Looks Like

R7$9vK!p3@tQ6#Lm

A truly strong password is long, random, and impossible to guess. It mixes uppercase and lowercase letters, numbers, and special characters in unpredictable combinations.

The good news? **You don't need to remember it.** Your password manager remembers it for you — instantly and securely.

THE FUTURE

Even Better: Passkeys

The future of login is **passkeys** — a passwordless technology that replaces traditional credentials with something you already have: **your body and your device**.

Fingerprint

Use your unique biometric print to verify identity

Face Recognition

A quick glance unlocks your account securely

Device Auth

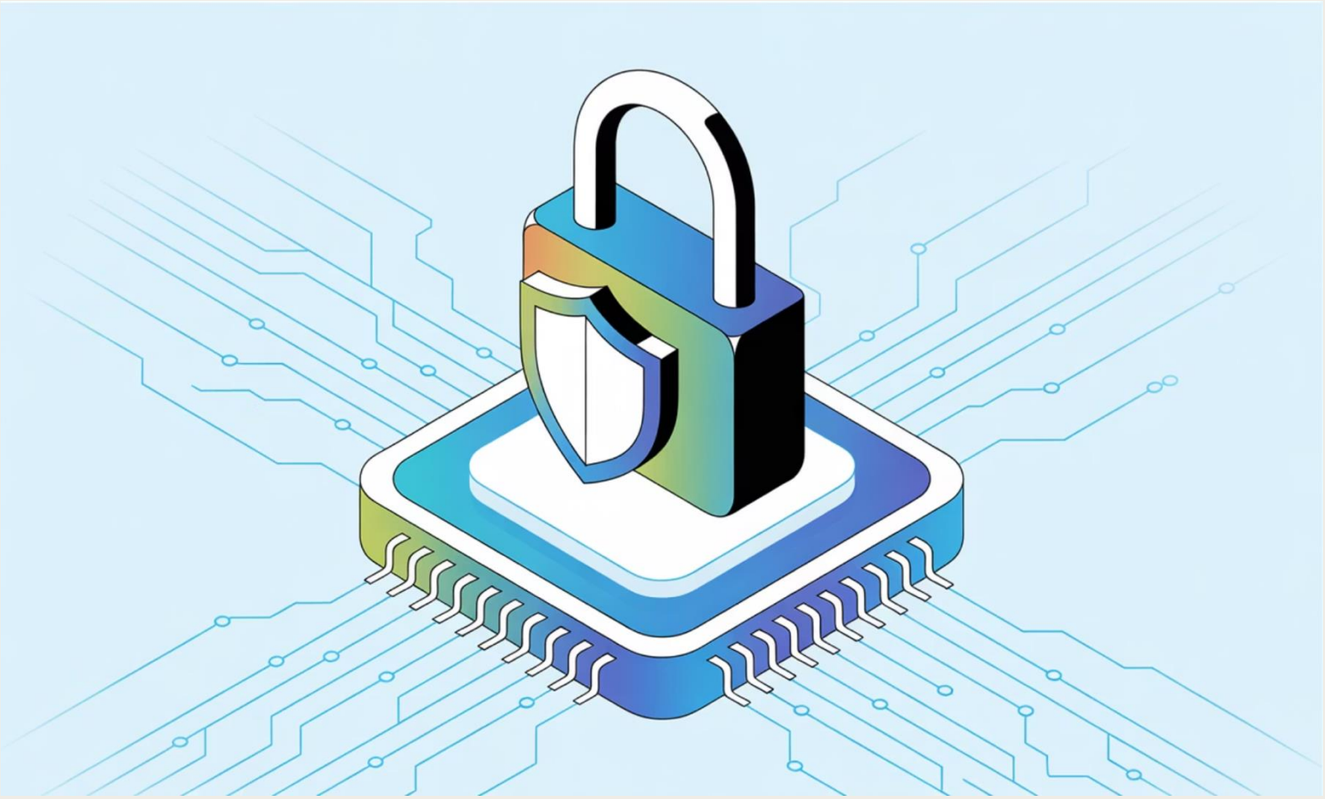
Your trusted device confirms it's really you

Already supported by **Apple**, **Google**, and **Microsoft** — and adoption is growing fast.

SECURITY

Why Passkeys Are Safer

Unlike traditional passwords, passkeys are cryptographically bound to **your specific device**. They never leave it, and they can't be intercepted in transit.



❌ Can't Be Guessed

No dictionary attacks or brute force can crack them

❌ Can't Be Stolen

Data breaches don't expose passkeys

❌ Can't Be Phished

Fake websites simply cannot capture them

HOW IT WORKS

Signing In — It's That Simple

Forget typing passwords. With passkeys, the entire login experience takes just seconds.

Tap "Sign in with passkey"

A single button replaces the password field

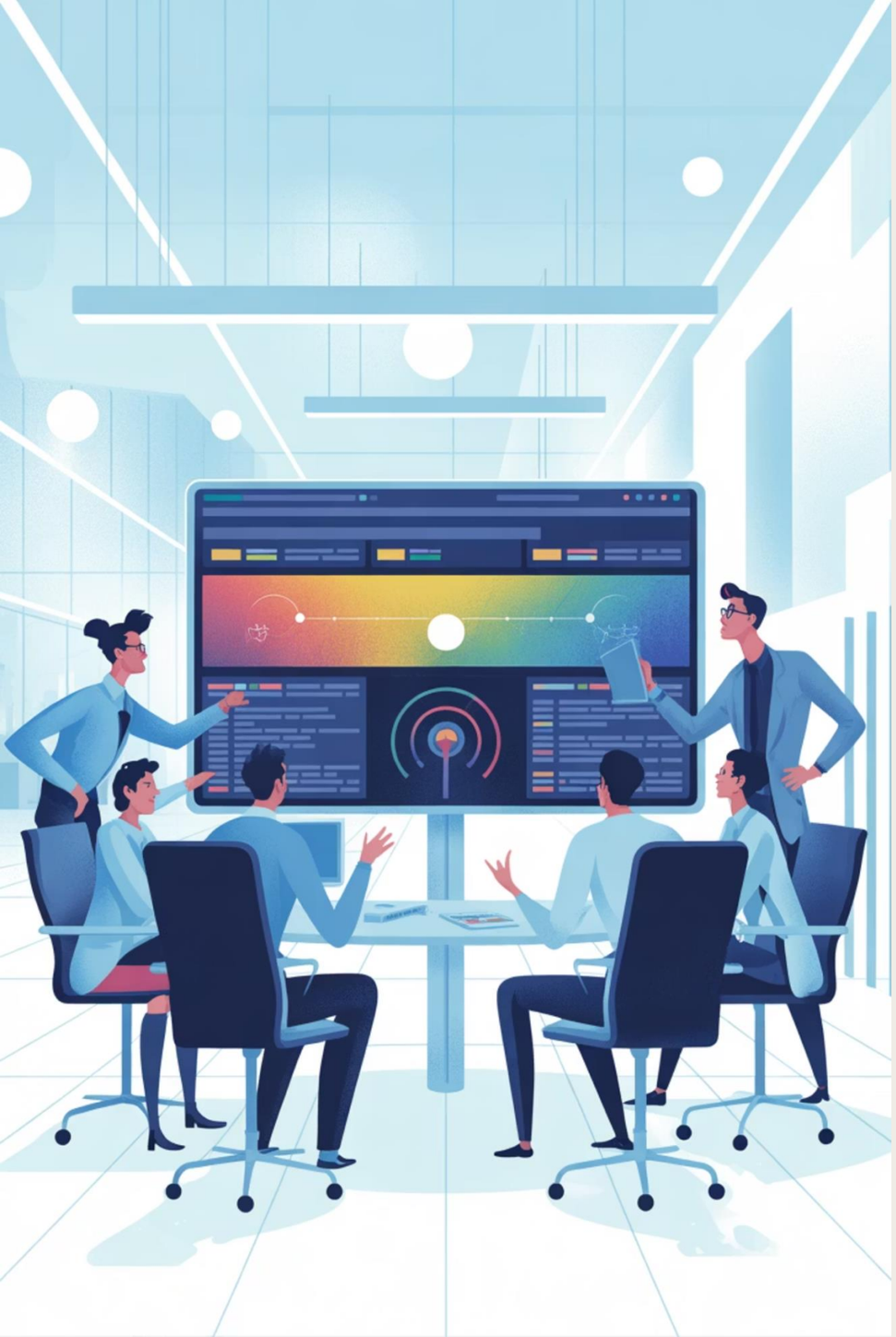
Face ID or Fingerprint

Your device verifies your identity biometrically

You're Logged In

Secure, instant, and completely passwordless

No passwords to type, forget, or reset — ever again.



ORGANIZATION

What Your Organization Should D

Implementing a few foundational security measures can **dramatically reduce risk** across your entire organization.

01

Deploy a Password Manager

Provide every employee with a managed, approved password vault

02

Enforce Unique Passwords

Require unique, randomly generated passwords for every service

03

Enable MFA Everywhere

Multi-factor authentication adds a critical second layer of defense

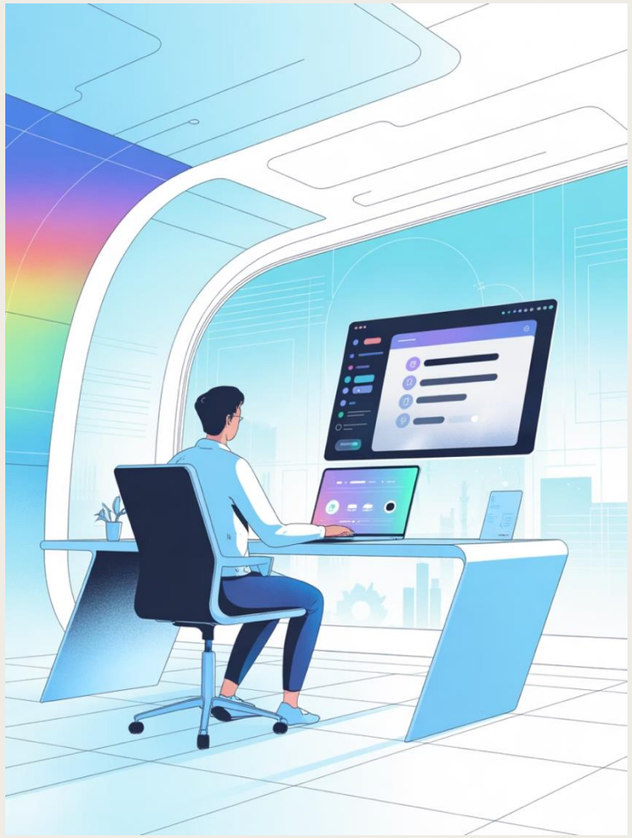
04

Adopt Passkeys

Begin rolling out passkey support for compatible platforms and apps

PERSONAL SAFETY

What Employees Should Do



Every individual plays a role in organizational security. These personal habits make a significant difference:

- **Never Reuse Passwords**
  One breach can cascade across all your accounts
- **Use a Password Manager**
  Let software handle the complexity so you don't have to
- **Avoid Unknown Login Links**
  Phishing emails are the #1 way attackers steal credentials
- **Enable MFA on Key Accounts**
  Especially email, banking, and work-related platforms

SELF-ASSESSMENT

Quick Security Checklist

Take 60 seconds to ask yourself these four questions. If the answer to any of them is "no" — **start today**.

1

Unique Passwords?

Is every one of your passwords different across all accounts?

2

Password Manager?

Are you using a dedicated tool to store and generate credentials?

3

MFA Enabled?

Have you turned on multi-factor authentication for critical logins?

4

Using Passkeys?

Are you taking advantage of passkeys wherever they're available?

☆ SUMMARY

Key Takeaways

1 Passwords Alone Aren't Enough

Traditional passwords are vulnerable to breaches, phishing, and brute force attacks

2 Password Managers Make Security Ea

They generate, store, and autofill strong credentials so you don't have to

3 Passkeys Are the Future

Biometric, device-bound authentication eliminates the biggest password risks

4 Small Changes Prevent Major Attacks

A few simple habits today can protect you and your organization tomorrow

PRESENTING TIPS

Making It Memorable for Your Audience

Tell a Story

"Imagine someone logging into your email and sending fake invoices to vendors — pretending to be you."

Real-world scenarios create urgency and emotional connection with your audience.

Do a Live Demo

Show your audience what modern security looks like in action:

- A password manager generating a strong password instantly
- Signing in with a passkey on your phone in seconds

Seeing is believing — and it makes the talk unforgettable.

Thank You

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Made with Gamma